A SMOOTH MORPHISM IS ETALE-LOCALLY A PROJECTION FROM AN AFFINE SPACE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Smoothness at a point is witnessed by a standard smooth presentation. Lemma \ref{lemma-smooth-locally-standard-smooth}, cited at proof lines 8562-8564. Provides the affine open and the presentation.
- The standard smooth presentation gives an etale morphism to an affine space. Algebra, Lemma \ref{algebra-lemma-standard-smooth} and its consequences, used at proof lines 8565-8598. Produces the etale morphism pi.

Source reference. Original lemma 054L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6644–6644. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01V7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2436–2436. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $\varphi : X \to Y$ be a morphism of schemes. Let $x \in X$. Let $V \subset Y$ be an affine open neighbourhood of $\varphi(x)$. If φ is smooth at x, then there exists an integer $d \geq 0$ and an affine open $U \subset X$ with $x \in U$ and $\varphi(U) \subset V$ such that there*

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exists a commutative diagram

$$\begin{array}{ccccc} X & \longleftarrow & U & \xrightarrow[\pi]{} & \mathbf{A}^d_V \\ \downarrow & & \downarrow & \swarrow & \\ Y & \longleftarrow & V & & \end{array}$$

where π is étale.

Proof. By Lemma 01V7 we can find an affine open U as in the lemma such that $\varphi|_U : U \to V$ is standard smooth. Write $U = \mathrm{Spec}(A)$ and $V = \mathrm{Spec}(R)$ so that we can write

$$A = R[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$$

with

$$g = \det \begin{pmatrix} \partial f_1/\partial x_1 & \partial f_2/\partial x_1 & \ldots & \partial f_c/\partial x_1 \\ \partial f_1/\partial x_2 & \partial f_2/\partial x_2 & \ldots & \partial f_c/\partial x_2 \\ \ldots & \ldots & \ldots & \ldots \\ \partial f_1/\partial x_c & \partial f_2/\partial x_c & \ldots & \partial f_c/\partial x_c \end{pmatrix}$$

mapping to an invertible element of A. Then it is clear that $R[x_{c+1}, \ldots, x_n] \to A$ is standard smooth of relative dimension 0. Hence it is smooth of relative dimension 0. In other words the ring map $R[x_{c+1}, \ldots, x_n] \to A$ is étale. As $\mathbf{A}^{n-c}_V = \mathrm{Spec}(R[x_{c+1}, \ldots, x_n])$ the lemma with $d = n - c$. □